A Cooperative Computation Offloading Strategy With On-Demand Deployment of Multi-UAVs in UAV-Aided Mobile Edge Computing

Chunlin Li, Yongzheng Gan, Yong Zhang, and Youlong Luo

***Abstract*—In this paper, we plan to use ground-based stations in mobile edge computing (MEC) and unmanned aerial vehicles (UAVs) to provide communication and computation offloading services in disaster areas. However, optimizing the initial number and three-dimensional position of deployed UAVs is a prerequisite for providing computing services to users. Additionally, due to the limited battery and computing power of UAVs, it is a major challenge to rationally design the UAV trajectory during the computational offloading period to ensure communication quality for mobile users and reduce the energy consumption for completing tasks. Thus, we propose a cooperative computation offloading strategy with on-demand deployment of multi-UAV in UAV-aided MEC. The strategy utilizes the predicted user trajectory for UAV deployment on the premise of the minimum path loss of users. Then, to minimize total energy consumption for completing tasks, a joint optimization problem comprising user association strategy, computing resource allocation strategy, and UAV trajectory is proposed, which is a mixed-integer nonlinear program (MINLP). Therefore, to find the suboptimal solution, we use the block coordinate descent method to solve the problem. Numerical results show that the proposed algorithm can efficiently reduce the path loss by up to 18.55% and the total energy consumption by 18.28% compared to the benchmarks.**

***Index Terms*—Mobile edge computing (MEC), unmanned aerial vehicles (UAVs), computation offloading, on-demand deployment.**

I. Introduction

With the rapid development of new applications like the large-scale Internet of Things (IoT) and network services, users have higher requirements for network service quality and low latency requests [1]. Particularly in virtual reality (VR) applications, the motion-to-photon (MTP) delay required for users to watch VR videos should be less than 20 ms, and more than 20 ms will cause motion sickness symptoms [2]. Although the computing power of terminal devices is becoming more and more powerful, it is still unable to do a large number of complex mathematical activities like rendering computational tasks for VR videos in a limited amount of time [3]. Moreover, due to the limited power and computing resources, terminal devices are also unable to handle complex tasks for long periods of time [4].

Mobile edge computing (MEC) started to arise and be used to address the aforementioned issues [5]. By installing computational servers, such as roadside units, base stations, etc., at the IoT edge, MEC allows terminal devices to offload tasks to their nearby edge servers, which can reduce the energy consumption and latency required by terminal devices to process the tasks [6], [7]. However, edge servers are usually statically deployed on the ground and may be damaged due to natural disasters [8]. Therefore, some researchers provide computing offload services for terminal devices by carrying edge servers on UAVs as mobile base stations to replace damaged servers [9], [10].

At present, the research focused on UAV-aided MEC is mainly UAV deployment and computation offloading. Determining the quantity of UAVs and their deployment positions is the goal of UAV deployment. One of the purposes of computation offloading is to effectively use the computing resources of the edge server to reduce the energy consumption of processing all terminal computing tasks under the time delay limit acceptable to the user terminal [11]. The above researches are not separate but closely combined. The quality of the UAV deployment scheme can directly affect the energy consumption of the computation offloading.

The main challenges associated with the deployment of UAVs are the number and location of UAVs deployed. The deployment method in [12], [13] minimized transmit power consumption or request latency by considering deploying the UAV at a horizontal location at a fixed height. Zhang and Ansari [14] deployed multiple UAVs in three-dimension (3D) space to reduce the number of deployed UAVs. Alzenad et al. [15] regarded the placement of UAVs as a problem of circle layout in the horizontal dimension and a problem of the smallest outer circle in the spatial dimension to complete UAV deployment. Donevski et al. [16] improved the average ratio of priority users by jointly optimizing

Manuscript received 22 March 2023; revised 30 July 2023; accepted 8 November 2023. Date of publication 15 November 2023; date of current version 15 April 2024. The work was supported by the National Natural Science Foundation of China (NSFC) under grants (No.62372344, No.62171330), Key Research and Development Plan of Hubei Province of China (No.2023BAB075). The associate editor coordinating the review of this article and approving it for publication was M. Sayit. *(Corresponding author: Chunlin Li.)*

Chunlin Li is with the School of Computer and Artificial intelligence, Wuhan University of Technology, Wuhan 430063, China, and also with the Science and Technology on Parallel and Distributed Processing Laboratory, National University of Defense Technology, Changsha 410073, Hunan, China (e-mail: chunlinli2020@163.com).

Yongzheng Gan and Yong Zhang are with the School of Computer and Artificial Intelligence, Wuhan University of Technology, Wuhan 430063, China.

Youlong Luo is with the School of Management, Wuhan University of Technology, Wuhan 430063, China.

Digital Object Identifier 10.1109/TNSM.2023.3332899

1932-4537 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

TABLE I
LIST OF COMPARISON OF UAV DEPLOYMENT AND COMPUTATION OFFLOADING MECHANISMS

Works	contents	Optimization goals	Number of UAVs	UAV deployment dimensions	User accurate location	UAV trajectory control	UAV deployment optimization
[12]	UAV deployment	minimize power consumption	fixed	2D	known	-	-
[13]	UAV deployment	minimize the average request delay	dynamic	2D	known	-	-
[14]	UAV deployment	maximize the throughput in the access link	fixed	3D	known	-	-
[15]	UAV deployment	maximize the number of covered users	fixed	3D	known	-	-
[16]	UAV deployment	maximize the average rate of users	fixed	3D	known	-	-
[17]	UAV deployment	maximize the number of serviced users	dynamic	3D	known	-	-
[18]	UAV deployment	maximize the number of serviced users	fixed	3D	known	-	-
[19]	Computation offloading	maximize the computation bits of the whole system	-	-	known	Yes	No
[20]	Computation offloading	minimize the execution delays and energy consumptions of tasks	-	-	known	Yes	No
[21]	Computation offloading	minimize the energy and latency of UAVs and edge servers	-	-	known	No	No
[22]	Computation offloading	maximize the offloading efficiency of UAVs	-	-	known	No	No
[23]	Computation offloading	minimize the energy consumption of all tasks accomplishing	-	-	known	Yes	No
[24]	Computation offloading	minimize the energy and latency of tasks	-	-	known	Yes	No
[25]	Computation offloading	minimize task completion time for UAVs	-	-	known	No	Yes
This work	Computation offloading	minimize the sum of energy consumption for all tasks processing	dynamic	3D	Unknown	Yes	Yes

the allocation of computational resources for each priority category to complete the deployment of UAV in 3D space. Akram et al. [17] aims to meet the actual network constraints while maximizing the use of the least number of UAVs to deploy UAVs. Chen et al. [18] proposed the use of the multi-particle swarm optimization approach to deploy UAVs and decomposed the UAV's position during deployment into two directions: altitude and level. The above studies are based on known user locations, but in disaster scenarios, user locations may be difficult to be obtained in time due to communication interruptions and user movement. In addition, the UAV deployment in [15], [16], [18] sets the number of UAVs to a fixed value, but UAVs should be deployed on-demand based on the system requirements.

Currently, research on computation offloading in UAV-aided MEC faces many challenges, including high energy consumption, high latency, and low offloading efficiency. Therefore, most of the research usually takes energy consumption, time delay as the optimization objective and solves by heuristic algorithms, or deep reinforcement learning [19], [20]. Zhu et al. [21] proposed a computational offloading strategy that focuses on energy and delay of edge servers, and used simulated annealing particle swarm optimization to get an optimal data allocation scheme. Dai et al. [22] considered the battery and computing power of UAVs to design a offloading scheme with the aim of improving offloading efficiency. The UAV position in [21], [22] was static, but since the UAV has limited energy and time to process the task, the UAV trajectory needs to be planned to reduce the energy consumption and time delay of the communication. Therefore, Xiong et al. [23] simultaneously optimized the task offloading, bit allocation during transmission, and UAV flight trajectory to reduce the task's overall energy consumption, and the problem was solved employing an overall alternative algorithm. Gao et al. [24] proposed a heuristic algorithm that aims to maximize the total throughput, and allow an end user's task to be offloaded to multiple UAVs. However, optimizing the number and location of UAV deployments before computation offloading was not considered in the above study. Then, Sun et al. [25] optimized the deployment of UAVs and designed an offloading algorithm to minimize the time required for all tasks processing.

Table I provides a detailed comparison on various aspects of our work and the studies mentioned above. Most of the current research on UAV deployment is based on the assumption that the precise location of all users is known. But this assumption is unrealistic and difficult to implement in the disaster area. And, they do not consider the importance of user mobility. When the location of the mobile user cannot be obtained in time, We are unable to accurately deploy UAVs according to user demands. Compared to existing computation offloading studies, we have considered the impact of the number and initial deployment location of UAVs on computing offloading, UAVs trajectory, and computational resource allocation in UAVs-assisted MEC networks, which was not fully considered in previous research. Therefore, to address the problem of collaborative computational offloading in multi-UAV-aided MEC, three challenges need to be addressed in this paper:

- How to complete the initial deployment of UAVs to meet the Quality of Service (QoS) of users based on their historical trajectories?
- How to plan UAV trajectories to cope with continuous user movement and thus reduce total network communication latency and energy consumption?
- How to design a suitable offloading strategy that allows tasks to be offloaded to a UAV or a ground base station

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

(GBS) in order to reduce the energy consumption of completing all tasks under a certain latency constraint?

Motivated by the above reason, a cooperative computation offloading strategy with on-demand deployment of multi-UAVs in UAV-aided MEC is proposed to minimize the sum of energy consumption for all tasks processing. The main contributions of our work are as follows:

1) Due to communication interruptions, the current location of mobile users may not be obtained in time, which leads that UAVs equipped with edge servers cannot be accurately deployed based on the user's information. Therefore, we propose an aerial base station deployment method based on user mobility and QoS. According to the user locations predicted from the user history trajectories, a UAV deployment model is constructed with the goal of minimizing system's path loss. Then, we design a new algorithm to obtain an approximate optimal solution, which can obtain the number and 3D location of UAVs. Compared with the existing schemes, the proposed method can significantly lower the deployed UAV number and the system's total path loss.
2) To minimize the sum of energy consumption of all tasks processing under the latency limitation of mobile terminal task processing, a computation offloading strategy based on ground base station and multi-UAV cooperation is proposed. The strategy aims to minimize the sum of task processing energy consumption by jointly optimizing the UAV trajectory, the user's offloading strategy, and the edge server's computing resource allocation strategy after determining the initial deployment of the UAVs. The design is a mixed-integer nonconvex problem. To get the solution, we apply the block coordinate descent method to solve the problem. Compared with the existing schemes, the proposed strategy can significantly lower the system's energy consumption.

The remainder of this paper is organized as follows. Section II introduces the UAVs dynamic deployment model and computation offloading model. Section III presents the problem solution and algorithm design. The experimental validation of the proposed algorithm is provided in Section IV. The conclusion is given in Section V.

II. System Model and Problem Formula

According to Fig. 1, The UAV-aided MEC system considered has N mobile users, U UAVs, and M ground base stations (GBS), where the UAVs are quadrotor UAVs and carry edge servers as mobile base stations. We define the set of users as $\mathcal{N} = \{1, 2, \ldots, N\}$, the set of GBS as $\mathcal{M} = \{1, 2, \ldots, M\}$, and the set of UAVs as $\mathcal{U} = \{1, 2, \ldots, U\}$. Then, let $ES = \{1, \ldots, M, \ldots, M + U\}$ denote the set of edge servers (GBS and UAVs). The non-orthgonal multiple access (NOMA) is used for communication between the UAV and users. The frequency-division multiple access (FDMA) is used for communication between the GBS and users. Different users use different frequency spectra for the access links when communicating with the UAV or GBS [27]. Finally, Table II lists the notations used in this paper.

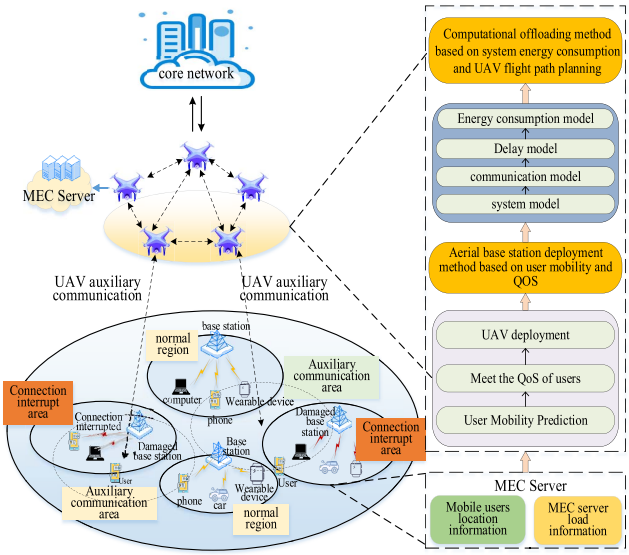


Fig. 1. System architecture of computation offloading strategy.

TABLE II
List of Main Notations

Notation	Description
$\mathcal{N}$	the set of mobile users
$\mathcal{M}$	the set of ground base stations (GBS)
$\mathcal{U}$	the set of UAVs
ES	the set of edge serve, including the GBS and UAVS
$\mathcal{K}$	the set of time slots in computation offloading period
$Track$	the user's history trajectory
st_{ij}	the probability of transferring state i to state j
P_{ij}^{LoS}	the LoS connection probability between user i and UAV j
L_{ij}^{NLoS}	the path loss of the LoS connection between user i and UAV j
L_{ij}	the average path loss between user i and UAV j
θ_{ij}	the elevation angle between user i and UAV j
ϕ_i	the maximum tolerable path loss for user i
r_{ij}	the horizontal distance between user i and UAV j
c_{ij}	whether the UAV j provides communication to the user i
$I_i(k)$	the task generated by user i within time slot k
F_i	the total number of CPU cycles required to execute the task $I_i(k)$
$a_{is}(k)$	the offload selection strategy for processing the task $I_i(k)$
$h_{ij}(k)$	the channel gain between UAV j and user i at time slot k
$R_{im}(k)$	the data transmission rate between user i and GBS m at time k
P_i	the transmission power of user i
W	the wireless channel bandwidth of the GBS m
$f_i^L(k)$	the computing resources used by user i to accomplish the task in time k
$T_{is}^{compute}(k)$	the time required to finish the task $I_i(k)$ in the edge server s
$f_{is}(k)$	the computational resources of the edge server s that user i can use in time slot k
$T_{is}(k)$	the overall execution time spent completing the task $I_i(k)$
$E_{ij}^{compute}(k)$	the energy consumption for edge server s to complete task the task $I_i(k)$

A. Aerial Base Station Deployment Method Based on User Mobility and QoS

1) User Mobility Prediction Model: Assuming that the user has been active in the disaster area for a period of time in the past, and the user's history trajectory is composed of a sequence of locations where the user is at each time point in the disaster area, denoted as $Track = \{l_0, l_1, \ldots, l_R\}$. And $l_r = (x_r, y_r, t_r), r \in \{0, 1, \ldots, R\}$ denotes the user is in the location (x_r, y_r) at time t_r. Then, we assume that the place where people stay more than $T_{\min}$ in a area of size $Row \times Col$ is hotspot area. Then, based on the user's

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

history trajectory, we can obtain the set of hotspot area $S:\{s_1,s_2,\ldots,s_H\}$.

The user location prediction in this paper focuses on the relationship between time and location. Therefore, one day are divided into ∂ time periods, each of which is $24/\partial$ (denoted as δ) hours. Then, the trajectory of a day is transformed into the vector v_d. And the user's trajectory within *D* days can be represented by the vector $V=\{v_1,\ldots,v_d,\ldots,v_D\}$.

Therefore, the hidden Markov model (HMM) can be established to complete the user's mobility prediction. we used $\lambda=\{ST,B,\pi\}$ to describe the HMM. Let $S:\{s_1,s_2,\ldots,s_H\}$ represent the *H* hidden states of the system, which represents the set of hotspots extracted from *Track*. Let $O:\{o_1,o_2,\ldots,o_{\partial\times D}\}$ denotes the observable states, which is the user's location in each time period, and $\mathbb{T}=\partial\times D$. Using $\pi=[P(s_1),P(s_2),\ldots,P(s_H)]$ to represent the initial state probability. And, $P(s_h)$ is the probability that the user is initially at s_h, which can be expressed as:

$$P(s_h)=\frac{Time(s_h,\mathbb{T})}{|\mathbb{T}|},\tag{1}$$

where $|\mathbb{T}|$ is total time spent by users in all hotspot area, and $Time(s_h,\mathbb{T})$ is the time the user stayed in hotspot s_h.

We assume that $ST=\{st_{ij}\}$ is a state transition matrix of $H\times H$, where $i,j\in\{1,2,\ldots,H\}$, and $st_{ij}=P(s_j^t|s_i^{t-1}.)$ denotes the probability of transferring state *i* to state *j*. Then, $P(s_j^t|s_i^{t-1}.)$ can be expressed as:

$$P\left(s_j^t\middle|s_i^{t-1}\right)=\frac{Time\left(s_i^{t-1}s_j^t,\mathbb{T}\right)}{Time\left(s_i^{t-1},\mathbb{T}\right)},\tag{2}$$

where $Time(s_i^{t-1},\mathbb{T})$ represents the ratio of the time that the user appears in the s_i in the time period $t-1$. $Time(s_i^{t-1}s_j^t,\mathbb{T})$ represents the product of the ratio of the time that the user appears in the hotspot area s_i in the time period $t-1$ and the ratio of the time that the user appears in the hotspot area s_i in the time period *t*.

Assuming that $B=\{b_{ij}\}$ is a confusion matrix of $H\times\partial$, where $i\in\{1,2,\ldots,H\},j\in\{1,2,\ldots,\partial\}$, and $b_{ij}=P(o_j^t|s_i^t.)$ denotes the probability that an observable symbol o_j is emitted from the hidden layer s_j. Then, we can obtain

$$P(o_j|s_i)=\frac{Time\left(s_i^{o_j},\mathbb{T}\right)}{Time(s_i,\mathbb{T})},\tag{3}$$

where $Time(s_i^{o_j},\mathbb{T})$ is the time the user spents in the hotspot area s_i before o_j, and $Time(s_i,\mathbb{T})$ is the time that the user spends in the hotspot area s_i.

2) Aerial Base Station Deployment Model: Assuming that the disaster area is a rectangular rectangle of $X\times Y$ with *N* users distributed. Then, the set of all user positions can be expressed as $UE=\{ue_i|ue_i=(x_i,y_i,0).\},i\in\mathcal{N}$, which can be obtained by HMM. Let $UAV=\{uav_j|uav_j=(x_j,y_j,z_j)\},j\in\mathcal{U}$ represent the set of UAVs positions for air-assisted communication.

Between the UAV and the user, we intend to use the Air-To-Ground (A2G) communication channel, which has line-of-sight (LoS) communication and non-line-of-sight (NLoS) communication [26]. The scene studied in this paper is a post-disaster scene in which there are damaged buildings, so there are both LoS and NLoS communication models. Therefore, the LoS connection probability between user *i* and UAV *j* is:

$$P_{ij}^{LoS}=\frac{1}{1+\alpha\exp\left(-\beta\left(\frac{180}{\pi}\theta_{ij}-\alpha\right)\right)},\tag{4}$$

where α and β are environment-dependent parameters, θ_{ij} is the elevation angle, so $\theta_{ij}=\arctan(z_j/r_{ij})$, where z_j is the height of UAV *j*, and r_{ij} is the horizontal distance.

Since there are only LoS connection and NLoS connection in A2G channels.Then, the probability of NLoS connection is

$$P_{ij}^{NLoS}=1-P_{ij}^{LoS}\tag{5}$$

In addition, the path loss of the LoS connection and the NLoS connection between user *i* and UAV *j* is as follows:

$$L_{ij}^{LoS}=20\log\left(\frac{4\pi f_c d_{ij}}{c_0}\right)+\eta_{LoS},\tag{6}$$

$$L_{ij}^{NLoS}=20\log\left(\frac{4\pi f_c d_{ij}}{c_0}\right)+\eta_{NLoS},\tag{7}$$

where η_{NLoS} and η_{LoS} denote the attenuation factors for the NLoS channel and the LoS channel, respectively, f_c is the carrier frequency, d_{ij} is the distance between UAV *j* and user *i*, and c_0 is the velocity of light. Then, the average path loss L_{ij} is as follows:

$$L_{ij}=P_{ij}^{LoS}L_{ij}^{LoS}+\left(1-P_{ij}^{LoS}\right)L_{ij}^{NLoS}.\tag{8}$$

By substituting formulas (4), (5), (6) and (7) into formula (8), and setting $\sigma_1=\eta_{LoS}-\eta_{NLoS}$, $\sigma_2=20log(4\pi f_c/c\,)+\eta_{NLoS}$, then L_{ij} also can be expressed as:

$$L_{ij}=20log\left(\frac{r_{ij}}{\cos(\theta_{ij})}\right)+\frac{\sigma_1}{1+\alpha exp\left(-\beta\left(\frac{180}{\pi}\theta_{ij}-\alpha\right)\right)}+\sigma_2.\tag{9}$$

In formula (9), for the parameters α and β in a specific environment, given the UAV height z_j, all ground points located at *r* from the UAV's horizontal center have the same path loss *L*. And with the UAV as the center, the horizontal distance between UAV and an additional nearby UAV is r'. So long as the region satisfies $r'\le r$, the path loss L' between UAV and an additional nearby UAV satisfies $L'\le L$.

By taking the partial derivative of the horizontal distance r_{ij} and elevation angle θ_{ij} in formula (9). Then, we can obtain

$$\frac{\partial L}{\partial r}=\frac{20}{r_{ij}\ln 2}\ge 0,\tag{10}$$

$$\frac{\partial L}{\partial\theta}=20\tan\theta_{ij}+\frac{\sigma_1}{\left(1+\alpha exp\left(-\beta\left(\frac{180}{\pi}\theta_{ij}-\alpha\right)\right)\right)^2}*\alpha exp\left(-\beta\left(\frac{180}{\pi}\theta_{ij}-\alpha\right)\right)*\frac{180\beta}{\pi}\ge 0.\tag{11}$$

In the formulas (10) and (11), when the user is closer to the UAV *j*, the horizontal distance r_{ij} becomes smaller and θ_{ij} becomes larger. However, the influence of the radius on $L(r_{ij},\theta_{ij})$ is higher than that of elevation angle θ_{ij}. Therefore, when r_{ij} reaches its maximum value r_{ij}^{max}, $L(r_{ij},\theta_{ij})$ will have a maximum value ϕ_i. When $L(r_{ij},\theta_{ij})=\phi_i$, r_{ij} can get

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

a maximum value. When $\partial r_{ij}/\partial\theta_{ij} = 0$, we can find the best elevation angle θ_{ij}^* at the maximum radius r_{ij}^{max}. And θ_{ij}^* can be obtained by the following:

$$\frac{\pi}{9\ln\ (10)} tan\theta_{ij}^* + \frac{\alpha\beta\sigma_1 exp\Big(-\beta(\frac{180}{\pi} arctan\theta_{ij}^* - \alpha)\Big)}{\Big(1+\alpha\exp(-\beta(\frac{180}{\pi}\theta_{ij}^* - \alpha))\Big)^2} = 0\,. \tag{12}$$

By bringing $\theta_{ij} = \theta_{ij}^*$ into $L(r_{ij}, \theta_{ij}) = \phi_i$, the maximum horizontal distance $r_{ij}^{\max}$, which can be expressed as:

$$r_{ij}^{\max} = \frac{c\cos\theta_{ij}^*}{4\pi f} 10^{\frac{\phi_i - P_{ij}^{Los}\eta_{Los} - P_{ij}^{NLos}\eta_{NLos}}{20}}. \tag{13}$$

Therefore, the optimal height z_{ij}^* is the UAV's height at which UAV *j* produces $r_{ij}^{\max}$ from user *i*, which is as follows:

$$z_{ij}^* = r_{ij}^{max}\tan\theta_{ij}^*. \tag{14}$$

The goal of the UAV deployment in this paper is to deploy a certain number of UAVsto minimize the total path loss for all users under the limit of user's QoS. Then, we assume that $c_{ij} = \{0,1\}, \forall i \in \mathcal{N}, j \in \mathcal{U}$ denote whether the UAV *j* provides communication to the user *i*. Therefore, the problem can be mathematically represented as follows:

$$\mathcal{N},\mathcal{U},c_{ij} \quad \min \sum_{i=1}^{N}\sum_{j=1}^{U} L_{ij}\,c_{ij} \tag{15a}$$

$$\text{s.t.} \quad \sum_{j=1}^{U} L_{ij}\,c_{ij} \le \phi_i, \forall i \in \mathcal{N}, \tag{15b}$$

$$\sum_{j=1}^{M} c_{ij} = 1, \forall i \in \mathcal{N}, \tag{15c}$$

$$c_{ij} = \{0,1\}, \forall i \in \mathcal{N}, j \in \mathcal{U}, \tag{15d}$$

$$0 \le x_j \le X, \forall j \in \mathcal{U}, \tag{15e}$$

$$0 \le y_j \le Y, \forall j \in \mathcal{U}, \tag{15f}$$

$$0 \le z_j \le h_{max}, \forall j \in \mathcal{U}. \tag{15g}$$

where constraint (15b) ensures that the UAV will be deployed to meet the QoS of each user. Constraints (15c) and (15d) guarantee that each user receives assistance from the appropriate UAV and that only one UAV will be providing assistance at the same time. Constraints (15e), (15f) and (15g) are the location constraints on UAV deployment in 3D.

B. Computation Offloading Model Based on Ground Base Station and Multi-UAV Cooperation

1) System Model: In this paper, we divide the offloading period into *K* time slots, denoted as $\mathcal{K} = \{1, 2, \ldots, K\}$. In each time slot, user *i* generates a task $I_i(k) = \{D_i(k), F_i(k)\}, \forall k \in \mathcal{K}$, where $D_i(k)$ denotes the data size of the task generated by the user *i* at the time *k*, and $F_i(k)$ denotes the total number of CPU cycles required to execute the task.

The distribution of *N* users is given by the HMM. Then, we denote $ue_i = \{ue_i(k)\}, \forall k \in \mathcal{K}$ is the trajectory of user *i* during offloading period,which consists of a sequence of the user's positions within each time slot. And $ue_i(k) = (x_i(k), y_i(k), 0)$ is the position of the user *i* in time slot *k*. The set of *M* available base ground stations (GBS) is $\mathcal{M} = \{1, 2, \ldots, M\}$. The position coordinates of GBS *m* is $(x_m, y_m, 0)$.

We assume that the UAV communicates with the user after each successive action. Because the action of the UAV is continuous, so we can consider it as real-time communication between the UAV and the user, therefore, the UAV trajectory is composed of a sequence of positions of the UAV after completing successive actions within each time slot [8]. The position of UAV *j* in the time slot *k* is $q_j(k) = (x_j(k), y_j(k), z_j)$. Therefore, within the offloading period, the trajectory of the UAV *j* is $q_j = \{q_j(k)\}, k \in \mathcal{K}$, and the trajectory of all UAVs is $Q = \{q_j\}, j \in \mathcal{U}$.

For the task $I_i(k)$, user *i* can choose to execute locally or select a edge serve from the set of edge servers $ES = \{1, \ldots, M, \ldots, M+U\}$ (including the GBS and UAVS) to offload the task $I_i(k)$. Then, we denote a_{ij} as the offloading selection strategy, which can be expressed as:

$$a_{is}(k) = \{0,1\}, \forall i \in \mathcal{N}, s \in \{0, 1, \ldots, M+U\}, k \in \mathcal{K} \tag{16}$$

where $a_{i0} = 1$ represents local execution, $a_{is}(k) = 1, s \in \{1, 2, \ldots, M\}$ represents that user *i* assigns the task to the GBS *m*, and $a_{is}(k) = 1, s \in \{M+1, M+2, \ldots, M+U\}$ represents that user offloads the task to UAV *j*. Therefore, let $A = \{a_{is}\}, \forall i \in \mathcal{N}, s \in \{0, 1, \ldots, M+U\}$ represents the selection strategy of *N* users in *K* time slots.

2) Communication Model: Users generally have three options when choosing tasks to process: local offloading, offloading to UAVs, and GBS.

When the task is offloaded to the UAV, the distance between UAV *j* and user *i* at time slot *k* is as follows:

$$d_{ij}(k) = \sqrt{\big(x_j(k) - x_i(k)\big)^2 + \big(y_j(k) - y_i(k)\big)^2 + z_j{}^2}. \tag{17}$$

Then, the channel gain between UAV *j* and user *i* at time slot *k* can be expressed as:

$$h_{ij}(k) = \frac{\rho_0}{d_{ij}(k)^2}, \tag{18}$$

where ρ_0 is the channel power gain at a reference distance of 1m. Then, the data transmission rate at time slot *k* is as follows:

$$R_{ij}(k) = B_0 \log\left(1 + \frac{P_i h_{ij}(k)}{\sigma^2}\right), \tag{19}$$

where B_0 is the communication bandwidth, P_i is the transmission power of user *i*, and σ^2 represents the power of Gaussian white noise.

Similarly, when the task is offloaded to the GBS *m*, the data transmission rate at time *k* is as follows:

$$R_{im}(k) = \frac{W}{Num} \log\left(1 + \frac{P_i h_{im}(k)}{\frac{W}{Num} N_0}\right), \tag{20}$$

where *W* is the wireless channel bandwidth of the GBS *m*, $h_{im}(k)$ is the channel gain, *Num* represents the number of users who choose to offload to the GBS *m* at time slot *k*, and N_0 is the noise power.

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

3) Delay Model: Since the results of computing tasks are generally very small, the downlink takes extremely little time compared to the uplink delay in the data backhaul process. Therefore, we only consider the task processing delay of edge servers and the uplink transmission delay of tasks.

At the time k, if task $I_i(k)$ is selected to be offloaded to the edge server s by user i, the total task completion time is $T_{is}^{upload}(k)$, which can be expressed as follows:

$$\begin{aligned} T_{is}^{upload}(k) &= T_{is}^{transfer}(k) + T_{is}^{compute}(k) \\ &= \frac{D_i(k)}{R_{is}(k)} + \frac{F_i(k)}{f_{is}(k)}, \end{aligned} \tag{21}$$

where $i \in \mathcal{N}, s \in ES, k \in \mathcal{K}$, $T_{is}^{transfer}(k)$ is the time spent by user i offloading the task $I_i(k)$ to the edge server s, and $T_{is}^{compute}(k)$ is the time required to finish the task $I_i(k)$ in the edge server s, and $f_{is}(k)$ is the computational resources of the edge server s that user i can use in time slot k.

In addition, due to the limited computational resources, there are the following constrains:

$$\sum_{i=1}^{N} a_{is}(k) f_{is}(k) \le f_s^{max}, \forall s \in ES, k \in \mathcal{K}. \tag{22}$$

where f_s^{max} is the edge server's maximum available computational resources.

Assuming that the local computing power consumption can be expressed as $k_i (f_i^L(k))^{v_i}$, k_i represents an effective switching capacitor, and the computing resources used by user i to accomplish the task in time k is $f_i^L(k)$. Therefore, the total time of local execution is as follows:

$$T_i^L(k) = \frac{F_i(k)}{f_i^L(k)}, \forall i \in \mathcal{N}, k \in \mathcal{K}. \tag{23}$$

Therefore, the time that the user i completes task $I_i(k)$ is as follows:

$$T_{is}(k) = \begin{cases} T_i^L(k) & \text{Local execution} \\ T_{is}^{upload}(k) & \text{Offloading to edge server } s. \end{cases} \tag{24}$$

Assuming that each task needs to be finished within T^{max} to guarantee the QoS of users, so we must meet the following constraints:

$$T_{is}(k) \le T^{max} \forall i \in \mathcal{N}, s \in \{0, 1, \ldots, M+U\}, k \in \mathcal{K}. \tag{25}$$

4) Energy Consumption Model: In this paper, there are two options for task computation: task local execution, and task offloading to edge servers s.

When the user chooses to execute the task locally, the energy consumption is as follows:

$$E_i^L(k) = k_i \left(f_i^L(k)\right)^{v_i} T_i^L(k), \forall i \in \mathcal{N}, k \in \mathcal{K}. \tag{26}$$

When the user chooses to offload the task $I_i(k)$ to the edge server s, the energy consumption is as follows:

$$\begin{aligned} E_{is}^{upload}(k) &= E_{is}^{transfer}(k) + E_{is}^{compute}(k) \\ &= P_i T_{is}^{transfer}(k) + k_s (f_{is}(k))^{v_s} T_{is}^{compute}(k). \end{aligned} \tag{27}$$

Thereby, the total energy consumption is as follows:

$$E = \sum_{i=1}^{N} \sum_{s=0}^{M+U} \sum_{k=1}^{K} a_{is}(k) E_{is}^{upload}(k) + (1 - a_{is}(k)) E_i^L(k). \tag{28}$$

5) Problem Formulation: In this paper, we only consider the energy consumption for processing all terminal tasks, which is mainly composed of three parts: the local calculation energy consumption when the user chooses to execute the task locally, the upload energy consumption when the user chooses to offload the task to the edge server s, and the calculation energy consumption when the edge server s calculates the task. Then, the goal of the proposed computational offloading method is to minimize the sum of energy consumption required to process all tasks. Additionally, within the offloading time, the initial takeoff position and final landing position of the UAV j are represented by q_{j0} and q_{jF}, where q_{j0} can be given by the deployment method of the UAV. And the UAV has a maximum speed V_{max} by considering its restricted kinetic energy. Therefore, the problem can be expressed as:

$$\min_{A,Q,F} \quad E \tag{29a}$$

$$\text{s.t.} \quad q_{j0} = q_j(1), \forall j \in \mathcal{U} \tag{29b}$$

$$q_{jF} = q_j(K), \forall j \in \mathcal{U} \tag{29c}$$

$$\left\| q_j(k+1) - q_j(k) \right\| \le \delta_k V_{max}, \forall j \in \mathcal{U}, \forall k \in \mathcal{K}, \tag{29d}$$

$$0 \le x_j(k) \le X, \forall j \in \mathcal{U}, \forall k \in \mathcal{K}, \tag{29e}$$

$$0 \le y_j(k) \le Y, \forall j \in \mathcal{U}, \forall k \in \mathcal{K}, \tag{29f}$$

$$\sum_{s=0}^{M+U} a_{is}(k) = 1, \forall i \in \mathcal{N}, k \in \mathcal{K}, \tag{29g}$$

$$a_{is}(k) = \{0, 1\}, \forall i \in \mathcal{N}, s \in \{0, \ldots, M+U\}, k \in \mathcal{K}, \tag{29h}$$

$$\sum_{i=1}^{N} a_{is}(k) f_{is}(k) \le f_s^{max}, \forall s \in \{1, \ldots, M+U\}, k \in \mathcal{K}, \tag{29i}$$

$$T_{is}(k) \le T^{max}, \forall i \in \mathcal{N}, s \in \{0, \ldots, M+U\}, k \in \mathcal{K}, \tag{29j}$$

where (29b) and (29c) give the initial takeoff position and final landing position of the UAV. Equation (29d) guarantes UAV's max speed in flight. Equations (29e) and (29f) ensure that the UAV can only fly in the specified rectangular area. Equations (29g) and (29h) ensure that each task is limited to being offloaded to a edge node at any time. Equation (29i) ensures that the edge node's compute resources are kept within its upper bound. Equation (29j) guarantees the QoS of the user during the offloading process.

III. Problem Solution and Algorithm Description

A. Outline of Proposed Solution

To solve problems (15) and (29), we propose Algorithms 1-4 to get the UAV deployment strategy and computation offloading strategy. First, based on the user's historical movement trajectory, we obtain the user's location in the future period by using Algorithm 1. Then, to obtain the

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 User Mobility Prediction Algorithm

Input: The user's history trajectory $Track = l_0, l_1, ..., l_R$
Output: The user's location O

Initialize $Row \times Col$, dwell time threshold $T_{\min}$, and prediction time interval δ
Extract the set of hotspots area S :'$\{s_1, ..., s_H\}$ from the user's history trajectory $Track = l_0, l_1, ..., l_R$
Convert the user's trajectory to a vector v_i
Initialize $\pi = [P(s_1), ..., P(s_H)]$, the state transition matrix A, and the confusion matrix B to establish HMM
Adjust model parameters to obtain optimal parameters λ
return user's location O

Algorithm 2 UAVs Deployment Algorithm

Input: Number of users N, disaster area $X \times Y$
Output: UAV deployment set UAV

Initialize user set UE according to Algorithm 1
Initialize U_1, U_2 and the number of UAVs $j = 0$
Initialize user 's path loss requirement set ϕ
while $U_1 \neq \emptyset$ **do**
  Find user i from U_1, and $i = \min_i \{\phi_i | i \in U_1\}$
  Calculate the maximum coverage radius r that meets the path loss requirements of user i
  $j = j + 1$, and calculate the height z_j of the UAV j
  for $(x_k, y_k) \in (x_k - y_i)^2 + (y_k - y_i)^2 = r^2$ **do**
    Find loc with the minimum total path loss
  end for
  Put the UAV with position loc and height z_j into UAV
  Associate users within the coverage radius r of UAV j
  Update the set of U_1 and U_2
end while

number of UAV deployments as well as the 3D location, we design a heuristic UAV deployment algorithm (Algorithm 2) to solve problem (15). After the initial UAV deployment, we use the block coordinate descent (BCD) method to divide the problem (29) into three subproblems: the user-edge node association strategy A, the edge node computing resource allocation strategy F and the UAV trajectory strategy Q, and solve these subproblems iteratively by Algorithm 4. Among them, we propose a heuristic greedy algorithm (Algorithm 3) to solve the subproblem (32). Finally, we give a convergence proof to verify that Algorithm 4 is feasible.

B. Aerial Base Station Deployment Algorithmic Based on User Mobility and QoS

For a given HMM $\lambda = \{ST, B, \pi\}$, the most likely position (hidden state) is found in each time slot δ(observable state) of the next day. Therefore, when predicting the mobility of users on day d, given the user's historical trajectory vector from day 1 to day $d-1$, the HMM updates the HMM regularly at the end of day $d-1$. This process is called parameter learning. To obtain better parameters to describe the system structure, the Baum-Welch algorithm is used to complete the parameter learning [28]. Then, the core pseudo code of the user mobility prediction algorithm based on HMM is shown in the Algorithm 1.

The Baum-Welch algorithm for HMM is an iterative optimization algorithm. In each iteration, the algorithm first calculates the forward and backward probabilities of the observed sequence, This phase's time complexity is $O(H^2\mathbb{T})$. Then it updates the parameters of the model, including the hidden state transfer probabilities and the observed state probabilities, and the time complexity is $O(H^2)$. Therefore, the overall time complexity of Algorithm 1 is $\mathrm{O}(LH^2\mathbb{T})$, where L is the number of iterations, H is the number of hidden states, $\mathbb{T}$ is the length of the observation sequence.

Since the application scenario in this paper is a post disaster scenario, it needs to make decisions quickly in a short time. Moreover, the heuristic algorithm can quickly find the problem's feasible solution and ignore the deviation among the feasible solution and the best solution. Then, we use a heuristic algorithm to solve the proposed problem.

The basic idea of the UAV optimal deployment method in this paper is to deploy the UAV iteratively in the affected area until all users' QoS demands are satisfied. In each iteration, under the premise of satisfying the user 's QoS, the UAV will be deployed in the appropriate location, where there is the least amount of path loss for all users connected to it.

Before UAV deployment, all users in the affected area are divided into two groups: U_1 and U_2. U_1 is the user set that has not been associated with the UAV, and U_2 is the user set that has been associated with the UAV. The initial states are $U_1 = UE$ and $U_1 = \emptyset$. After each round of deployment, the UAV will deploy its horizontal position at loc, with the height of z, and all users belonging to the UAV association area will be associated. After user association, U_1 and U_2 will be updated. Iterative calculation until each user makes connections to the UAV. Then, the deployment of UAVs is completed.

For the circle O_i where r is the radius and the user i serves as the center, the set of O_i can be expressed as $O_i = \{(x_{oi}, y_{oi}) | (x_{oi} - x_i)^2 + (y_{oi} - y_i)^2 = r^2\}$, and for the user set U_3 that has not been associated with the UAV, it can be expressed as $U_3 = \{(x_i, y_i) | (x_{oi} - x_i)^2 + (x_{oi} - y_i)^2 \leq r^2 \cap U_1\}$. Therefore, the total path loss L_{sum} between U_3 and UAV j is as follows:

$$L_{sum} = \sum_{i \in U_3} L_{ij}. \tag{30}$$

Therefore, the horizontal deployment position of UAV j is:

$$loc = \min_{L_{sum}} \{(x_j, y_j) \Big| (x_j - x_i)^2 + (y_j - y_i)^2 = r^2\}. \tag{31}$$

Then, the pseudo code of the UAVs deployment algorithm is displayed in Algorithm 2.

Algorithm 2 iteratively deploys UAVs in the affected area, determining the number of UAVs and the 3D deployment locations. In the process of deploying a UAV, the user with minimum path loss is first found in U1, which is a step with time complexity $O(N)$. Then, the 3D location of the UAV is determined based on this user's location and the users within range are associated with this UAV, a step with time complexity in $O(1)$. Thus, the overall time complexity of

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2 is $O(NU)$, where N is the number of users and U is the number of UAVs deployed.

C. Computation Offloading Algorithm Based on Ground Base Station and Multi-UAV Cooperation

Problem (29) is a mixed integer nonconvex problem. Therefore, we use the BCD method to divide the problem into three sub-problems: the user-edge node association strategy A, the edge node computing resource allocation strategy F and UAV trajectory strategy Q. By iteratively resolving the aforementioned three subproblems to find the problem's suboptimal solution.

1) Optimization of User-Edge Node Association Strategy: Given the edge server's computing resource allocation F and the UAV trajectory Q, problem (29) can be optimized as:

$$\min_{A} \quad E \tag{32a}$$

$$s.t. \quad \sum_{s=0}^{M+U} a_{is}(k) = 1, \tag{32b}$$

$$a_{is}(k) = \{0, 1\}, \tag{32c}$$

$$\sum_{i=1}^{N} a_{is}(k) f_{is}(k) \le f_s^{max}. \tag{32d}$$

To solve the above binary integer programming problem, we propose a heuristic algorithm. The purpose of this algorithm is to minimize the total energy consumption required to process tasks within a time slot based on the user's access order, allowing user i to selfishly select tasks to be executed locally or offloaded to edge server s in time slot k. By calculating the optimal user's offloading strategy in each time slot, the optimal solution of the whole problem is approximately obtained. The pseudo code of the user offloading decision algorithm is displayed in the Algorithm 3.

2) Optimization of Edge Node Computing Resource Allocation Strategy: Under the fixed user decision A and UAV flight trajectory Q, it is clear how much energy the user uses while choosing the local offloading part. Therefore, problem (29) can be optimized as:

$$E = \sum_{i=1}^{N} \sum_{s=1}^{M+U} \sum_{k=1}^{K} a_{is}(k) E_{is}^{upload}(k). \tag{33}$$

Energy consumption in (33) includes the upload energy consumption and the calculated energy consumption of the task at the edge server. By uploading the energy consumption expression, we find that there is no edge server computing resource variable F in this part, so this part is treated as a fixed value and the value is set to 3 [30]. Therefore, the objective function E could be optimized as follows:

$$\min_{F} \quad \sum_{i=1}^{N} \sum_{s=1}^{M+U} \sum_{k=1}^{K} a_{is}(k) \{k_s [f_{is}(k)]^2 F_i(k)\} \tag{34a}$$

$$\sum_{i=1}^{N} a_{is}(k) f_{is}(k) \le f_s^{max}, \tag{34b}$$

$$T_{is}(k) \le T^{max}. \tag{34c}$$

Algorithm 3 User Offloading Decision Algorithm

Input: Q, F, f_s^{max}, $I_i(k)$
Output: A

```
for k ∈ 𝒦 do
    Set user's optimal selection index = 0
    Set energy consumption E_k = 0
    for i ∈ 𝒩 do
        Calculate energy consumption e for local offloading
        for j ∈ {1, ..., M + U} do
            Calculate energy consumption e_ij between user i and edge serverj
            if e_ij < e and satisfy the constraint (29i) then
                a_iiindex(k) = 0
                Update computing resource of edge server index
                a_ij(k) = 1
                Update computing resource of edge server j
                index = j
            end if
        end for
        Update a_iiindex(k) = 1 to user offloading strategy A
        i = i + 1
    end for
    k = k + 1
end for
```

Then, taking formulas (20), (22), and (23) into (34c) for expansion can obtain following constraint:

$$f_{is}(k) \ge \frac{F_i(k)}{T^{\max} - \frac{D_i(k)}{B_0 \log_2\left(1 + P_i \rho_0 / \left(\sigma^2 \| q_j(k) - u e_i(k) \|^2\right)\right)}}. \tag{35}$$

The constraint (35) expresses that the computational resources provided by the edge server s to user i have a minimum value at time slot k. The user decision A and UAV flight trajectory Q have been determined, so the user that the server s needs to serve at time k has been determined. Assuming that $N_s(k)$ users need to be served, the meaning of this problem is that when server j serves $N_s(k)$ users, how to allocate available computing resources f_s^{max} to minimize E. Therefore, this problem could be solved by solving (36) to obtain the computing resource allocation of the server j at k.

$$\begin{bmatrix} k_s F_1(k) & -k_s F_2(k) & 0 & \cdots & 0 \\ 0 & k_s F_2(k) & -k_s F_3(k) & \cdots & 0 \\ 0 & 0 & k_s F_3(k) & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ -k_s F_1(k) & 0 & 0 & \cdots & k_s F_{N_s}(k) \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ \cdots \\ f_{N_s(k)} \end{bmatrix} = 0. \tag{36}$$

3) Optimization of Air Base Station Trajectory Strategy: Under fixed A and F, the energy consumption that users choose to offload to local and the edge server's computing energy consumption is fixed. Therefore, problem (29) can be optimized as:

$$\min_{Q} \quad \sum_{i=1}^{N} \sum_{s=1}^{M+U} \sum_{k=1}^{K} a_{is}(k) \frac{P_i D_i(k)}{B_0 \log_2\left(1 + \frac{P_i \rho_0}{\sigma^2 \| q_j(k) - u e_i(k) \|^2}\right)} \tag{37a}$$

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

$$\text{s.t.} \quad (29b)-(29f), \tag{37b}$$

$$\frac{D_i(k)}{B_0\log_2\left(1+\frac{P_i\rho_0}{\sigma^2\|q_j(k)-ue_i(k)\|^2}\right)}+\frac{F_i(k)}{f_{is}(k)}\le T^{\max}. \tag{37c}$$

Due to the existence of constraints (29d) and (37c), this subproblem is a non-convex problem. Then, we use Succcessive Convex Approximation (SCA) to deal with the problem. The steps of SCA method are to get the coordinate points of some air base station attachments given in the process of solving the UAV trajectory sub-problem each time. Through the properties of convex functions, the non-convex constraints in the trajectory problem of the air base station are transformed into convex constraints through Taylor formula expansion, and then the convex problem after approximate transformation is solved. Therefore, we first introduce set $\xi = \{\xi_{is}(k), \forall i \in \mathcal{N}, s \in \{1,2,\ldots,M+U\}\}$, and the formula of $\xi_{is}(k)$ is

$$\xi_{is}(k)=\frac{a_{is}(k)P_iD_i(k)}{B_0\log_2\left(1+\frac{P_i\rho_0}{\sigma^2\|q_j(k)-ue_i(k)\|^2}\right)}. \tag{38}$$

Then we use SCA method to solve problem. For any given local locus $Q_j^r(k)$, where $Q^r = \{Q_j^r(k), \forall j \in U, k \in K\}$, we obtain the following lower bound by Taylor expansion.

$$B_0\log_2\left(1+\frac{P_i\rho_0}{\sigma^2\left\|q_j(k)-ue_i(k)\right\|^2}\right)\ge y_j\left(\{Q_j^r(k)\}\right), \tag{39}$$

where the calculation formula of $y_j(\{Q_j^r(k)\})$ and $\gamma_j(k)$ is

$$y_j\left(\{Q_j^r(k)\}\right)=B_0\log_2\left(1+\frac{P_i\rho_0}{\sigma^2\left\|Q_j^r(k)-ue_i(k)\right\|^2}\right)+\gamma_j(k)\left(\|q_j(k)-ue_i(k)\|^2-\|Q_j^r(k)-ue_i(k)\|^2\right), \tag{40}$$

$$\gamma_j(k)=\frac{-\rho_0B_0P_i\log_2 e}{\sigma^2\left(\left\|Q_j^r(k)-ue_i(k)\right\|^2\right)\left(\frac{\|Q_j^r(k)-ue_i(k)\|^2\sigma^2+P_i\rho_0}{\sigma^2}\right)}. \tag{41}$$

Therefore, problem (29) could be transformed as follows:

$$\min_{Q,\xi}\ \sum_{i=1}^{N}\sum_{j=1}^{U}\sum_{k=1}^{K}\xi_{ij}(k) \tag{42a}$$

$$\text{s.t.} \quad (29b)-(29f), \tag{42b}$$

$$y_j\left(\{Q_j^r(k)\}\right)\ge\frac{a_{ij}(k)P_iD_i(k)}{\xi_{ij}(k)}, \tag{42c}$$

$$y_j\left(\{Q_j^r(k)\}\right)\ge\frac{D_i(k)}{T^{\max}-F_i(k)/f_{ij}(k)}. \tag{42d}$$

This subproblem is a convex quadratic constraint problem that can be solved using CVXPY solver [30].

4) Overall Algorithm: BCD algorithm can be broken down into several smaller problems to solve. Each time, the subproblem of one variable is optimized and the other variables remain unchanged. Then, by alternatively optimizing the subproblems of variable blocks, we can get the problem's solution.

Algorithm 4 Computation Offloading Algorithm

Input: M, U, N, K, $I_i(k)$, P_i, f_s^{max}, iterations $L=0$, maximum iterations $L_{\max}$, threshold θ

Output: E, A, Q, F

1: Initialize N user and U UAV positions according to Algorithm 1 and Algorithm 2
2: Initialize Q^0, A^0, F^0, and $E^0=0$
3: **while** $\left|E^{L+1}-E^L\right|\ge\theta$ and $L\le L_{\max}$ **do**
4: Given F^L and Q^L, find A^{L+1} based on Algorithm 3
5: Given A^{L+1} and Q^L, find F^{L+1} by solving (36)
6: Given A^{L+1} and F^{L+1}, find Q^{L+1} by using cvx
7: Calculate E^{L+1} according to A^{L+1}, Q^{L+1} and F^{L+1}
8: $L=L+1$
9: **end while**

Therefore, by using the above ideas, we divided the original problem into three sub-problems to solve. Then, the pseudo code of the computation offloading algorithm is displayed in the Algorithm 4.

5) Convergence and Complexity Analysis: For Algorithm 4, we need to prove that the target value in each iteration process of solving (29) is a non-increasing sequence and can converge to a feasible solution. In step 4 of Algorithm 4, the association problem (32) between the user and the edge server is solved by Algorithm 3. Similar to [31], [32], after $L+1$ iterations, by using the algorithm 3, we can obtain

$$E\left(A^{L+1},F^L,Q^L\right)\le E\left(A^L,F^L,Q^L\right). \tag{43}$$

The resource allocation policy of edge server s in time slot k is obtained by solving (36) in step 5, and then the optimal computational resource allocation policy for problem (34) can be obtained by computing (M+U) edge servers in K time slots. Thus, we can obtain

$$E\left(A^{L+1},F^{L+1},Q^L\right)\le E\left(A^{L+1},F^L,Q^L\right) \tag{44}$$

Let $E_{trj}^{lb,L}(A,F,Q)=E_{trj}^L$, where E_{trj}^L is the lower bound obtained by solving the UAV trajectory problem (42) in iteration L. Due to SCA technology, an effective approximate solution can be obtained and convergence to a local optimal solution can be guaranteed. Therefore, we can obtain

$$\begin{aligned}E\left(A^{L+1},F^{L+1},Q^L\right)&\overset{(a)}{=}E_{trj}^{lb,L}\left(A^{L+1},F^{L+1},Q^L\right)\\&\overset{(b)}{\ge}E_{trj}^{lb,L}\left(A^{L+1},F^{L+1},Q^{L+1}\right)\\&\overset{(c)}{\ge}E\left(A^{L+1},F^{L+1},Q^{L+1}\right)\end{aligned} \tag{45}$$

where (a) holds because the first-order Taylor expansion of problem (37) is compact in the given local point, which implies that (42) has the same objective value as problem (37) at Q^L; (b) holds because the optimal solution of problem (42) can be solved by step 6 of Algorithm 4; (c) holds because problem (42) uses the SCA method, and then the objective

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

value is a lower bound of the original problem (37) in Q^{L+1}. By using the results proved above, we can obtain

$$E\left(A^L, F^L, Q^L\right) \geq E\left(A^{L+1}, F^{L+1}, Q^{L+1}\right) \quad (46)$$

The above reasoning indicates that the target value of problem (29) is non increasing in each iteration and is finitely lower-bounded by zero. Therefore, Algorithm 4 converges.

The time complexity of algorithm 4 mainly comes from three aspects: First, Algorithm 1 is used to solve the subproblem (32). The time complexity is $O(K(N+U))$, where K is the number of time slots, N is the number of users, M is the number of GBS, and U is the number of UAVs. Second, to solve subproblem (34), the optimal allocation strategy of computing resources on the edge server s at time slot k can be obtained by solving (36). Then, the time complexity is $O(K(M+U))$. Third, we use the SCA method and the interior point method in CVX solver to solve the subproblem (37). Therefore, the time complexity is $O(L_1(KU)^{3.5})$, where L_1 is the number of iterations required by the CVX solver. Then, assuming that Algorithm 4 converges after L_2 iterations, the overall time complexity of Algorithm 4 is $O(L_2(KN(M+U)+K(M+U)+L_1(KU)^{3.5}))$, which means that Algorithm 4 can be efficiently solved with a polynomial time complexity.

Compared with benchmark algorithms, we find that the time complexity in [23] is $O(L(KNU+KN+L_2(KU)^{3.5}))$. The time complexity in [24] is $O(gn(l+m))$, where g is the number of iterations, n is the number of individuals, l is the length of chromosomes, and m is the number of mutation operations. The time complexity in [25] is $O(L_2(m^{2n}+L_1(2U)^{3.5}))$. Therefore, the time complexity of the proposed algorithm is roughly the same as [23], [25].

IV. Experimental Validation

A. Experimental Configuration

In this section, Python 3.7 is installed on an Intel Core i7 CPU computer with a processor speed of 2.4 GHz and 16 GB RAM to simulate and implement the two proposed algorithms. PyCharm2020 is installed on Windows 10 as the experimental simulation platform and XTDrone as the general simulation platform to simulate the flight path of UAVs. And, CVXPY solver is used to solve the problem of UAV trajectory. In the process of computation offloading, the task that users offload is face data. Then convolutional neural network (CNN) is applied to the process of face recognition.

1) Aerial Base Station Deployment Experiment: To evaluate the viability of the aerial base station deployment method based on user mobility and QoS, this paper uses GeoLife dataset [33] to evaluate the proposed prediction algorithm based on hidden Markov model. A total of 17621 movement tracks from 182 users are collected in the GeoLife dataset. Each track includes longitude, latitude, altitude and timestamp.

In this experiment, the size of disaster area ranges from $600 \times 600m^2$ to $1200 \times 1200m^2$. In addition, Table III provides some other experimental parameters for aerial base station deployment. Then, we select the number of UAVs, total throughput and total system path loss as the algorithm performance evaluation indicators.

TABLE III
Aerial Base Station Deployment Experimental Parameters

Parameter	Value
The number of time slice ∂	48
Environmental parameters α	9.61
Environmental parameters β	0.16
LOS channel link attenuation factor η_{los}	1 dB
NLOS channel link attenuation factor η_{nlos}	20 dB
Carrier frequency f_c	2 GHz
Error range	20 min
the number of mobile users N	[400,1000]
the user threshold	[105,135] dB
the available bandwidth of the UAV	[10,40] MHz

TABLE IV
Computation Offloading Experimental Parameters

Parameter	Value
Time period	2 min
Time interval δ_k	10 s
Number of users	100
Number of ground base stations	1
Scope of disaster area	$1 \times 1km^2$
Task collection size	[400,600]KB
Calculate CPU revolutions	1000 r/bit
User transmission power P	0.1 W
Noise power σ^2	-120 dBm
Channel power gain ρ_0	-50 dB
UAV bandwidth capacity B_0	50 MHz
Ground base station bandwidth capacity W	40 MHz
Maximum computing resources of UAVs f_{uav}^{max}	4 GHz
Maximum computing resources of GBS f_{EC}^{max}	3.5 GHz
Computing resources of User terminal $f_{UE}^{L}(k)$	2.5 GHZ

- *The number of UAVs:* The number of UAVs deployed should be sufficient to ensure the QoS including all users.
- *The total throughput:* The total number of bits per unit time that all users transmit data to the edge server.
- *The total path loss:* The loss of all users in the communication process in UAV-aided MEC.

To analyze the performance of the aerial base station deployment method based on user mobility and QoS proposed in this paper, we take ITA [16], SPUA [17], and MPSO [18] as comparison algorithms.

2) Computation Offloading Strategy Experiment: In the experimental verification of the computation offloading algorithm, we use the Labeled Faces in the Wild dataset [34]. The 13323 online photos in this dataset include 5749 celebrities, which can be used to simulate situations where ground base stations may be destroyed and assist in finding missing individuals through facial recognition.

The main experimental parameters of computation offloading strategy based on ground base station and multi-UAV cooperation are provided in Table IV. To evaluate the viability of the algorithm of computing offload strategy based on ground base station and multi-UAV cooperation, the total amount of data transmitted by the system, energy consumption and task delay are used as evaluation indicators in this paper.

- *The total amount of data transmitted by the system:* The total sum of the data transmitted by the terminal when the task is offloaded to the edge server.
- *The energy consumption:* It includes local computing energy consumption, task upload energy consumption, and edge server computing energy consumption.

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

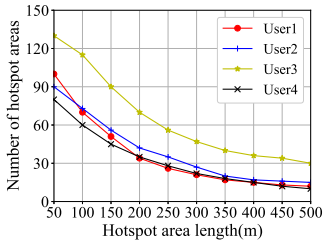


Fig. 2. Effect of region length on the number of hotspots.

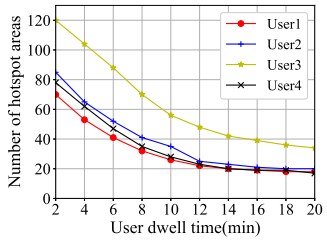


Fig. 3. Effect of dwell time on the number of hotspots.

- *The task delay:* The sum of the time required from the beginning to the completion of the task each time the user offloads the task from the system.

To analyze the performance of the computation offloading strategy based on ground base stations and multi-UAV cooperation in this paper, three other computational offload technique algorithms—TOA [23], HJPQ [24], and MUT [25] are contrasted with the proposed algorithm.

B. Results and Analysis

1) Experiment Verification of Air Base Station Deployment: Fig. 2 verifies the relationshipbetween the length of the hotspot area and the number of hotspot areas on four user tracks. In this experiment, the user dwell time of the proposed algorithm is set at 10 minutes. The area length range is constrained to 50 to 500 m. The data shows that as an area's length increases, the number of hotspots decreases, initially quickly and then gradually. And the data indicates that when the area length reaches 250 m, the length is at an inflection point. Therefore, this paper takes this length as the best area length.

The relationship between the dwell time $T_{\min}$ of four user's history trajectories and the number of hotspots is shown in Fig. 3. In this experiment, the length of area is set to 250 m. From Fig. 3, as the user's dwell time increases, the number of hotspots decreases gently and finally flattens. This is because as the user's dwell time increases, there will be more hotspots that do not meet the definition requirements. And when the dwell time is set to 12 minutes, it is most consistent with the user's dwell time setting in this paper.

Fig. 4 describes the influence of the time threshold of the user mobility prediction algorithm on prediction accuracy. The ratio of the accurate position prediction of the model to all forecast times is known as prediction accuracy. Assumed to be the temporal error threshold is τ. In time T, the prediction is accurate if, within the temporal error threshold τ, the anticipated position matches the actual position of the test

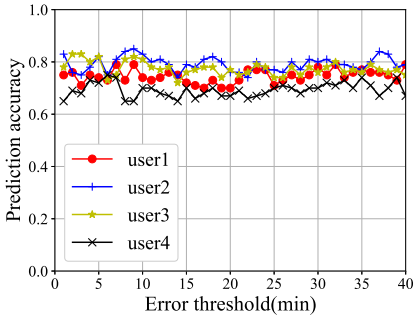


Fig. 4. Effect of time threshold on prediction accuracy.

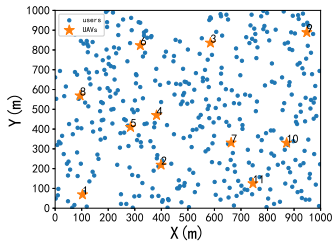


Fig. 5. Horizontal location.

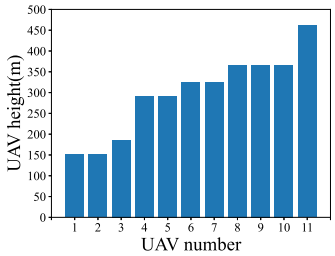


Fig. 6. Height of UAVs.

user. From Fig. 4, the forecast accuracy of the four test users' locations by using HMM is around 70%, which indicates that the prediction method in this paper is rather dependable.

After completing the experiment of user mobility prediction based on HMM, the location of all users can be obtained. We randomly selected 400 users and mapped their locations to the area of $1000 \times 1000\ m^2$. Then, we complete the optimal deployment of UAV based on user's QoS to get the optimal number of UAV deployments and 3D positions by using Algorithm 2. The optimal deployment scheme of UAV is depicted in Fig. 5 and Fig. 6. Fig. 5 shows that there are 11 UAVs deployed to meet the needs of 400 users in the area, and the two-dimensional plane positions and heights of each UAV deployment are shown in Fig. 5 and Fig. 6, respectively.

In Fig. 7(a), as more users in the disaster area, there are also more UAVs. This is because when the area of the affected area is constant, due to the restricted coverage provided by UAVs, more UAVs are needed to cover all users. Through further observation, the proposed algorithm uses the least number of UAVs. This is because, when planning the three-dimensional coordinates of the UAV, the proposed algorithm takes into account the users with the lowest user threshold among the remaining uncovered users each time, aiming to maximize the coverage of the users under the limit

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

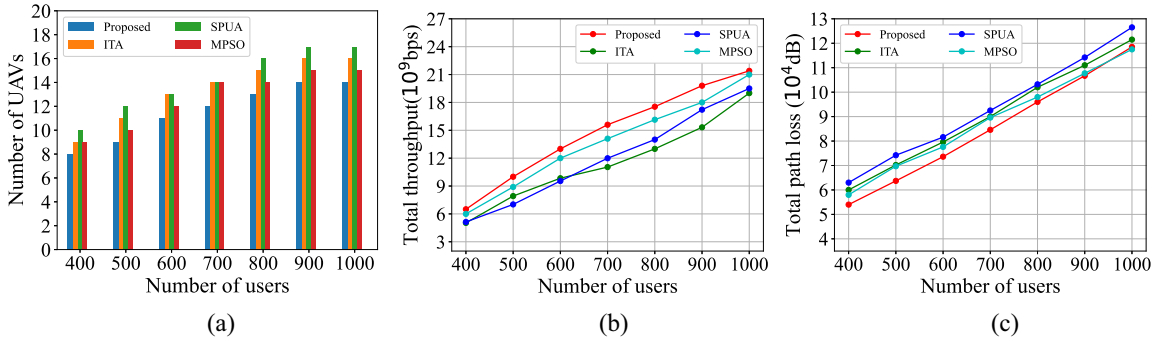


Fig. 7. Effect of the number of end users on experimental results. (a) The number of UAVs. (b) The total throughput. (c) The total path loss.

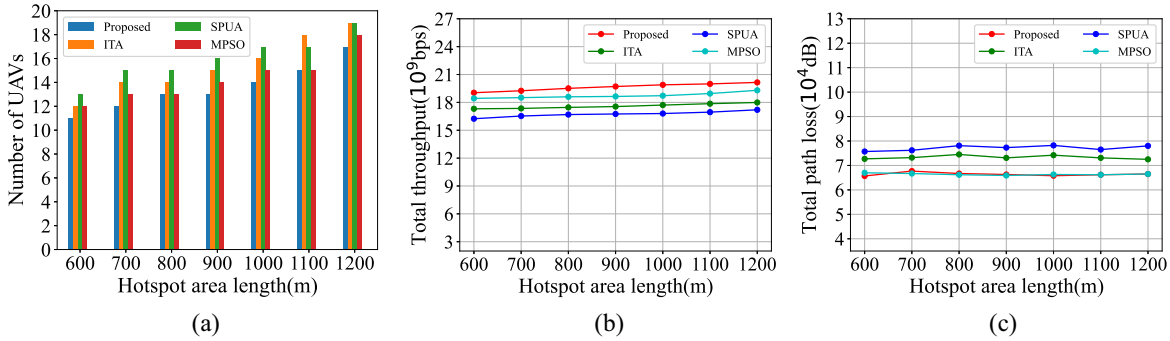


Fig. 8. Effect of the hotspot area length on experimental results. (a) The number of UAVs. (b) The total throughput. (c) The total path loss.

of user's QoS. In Fig. 7(b), as the number of users rises, so does the total throughput of the terminal equipment in the area. This is because as the number of users rises, more data is transmitted, exceeding the communication capacity of some UAVs and thereby reducing the throughput of the entire system. As shown in Fig. 7(c), with more users in the disaster area, the overall path loss rises. This is because the UAV deployment strategy of the ITA algorithm wastes some communication resources of UAVs, resulting in the need to deploy additional UAVs. The SPUA algorithm uses the UAV video acquisition method to obtain the user's location, which has certain inaccuracies. The MPSO algorithm obtains an optimal coverage position that satisfies the termination condition, its efficiency in terms of both the number of UAVs and throughput is basically the same as that of the proposed algorithm. When the number of users is 700, the total path loss of the proposed algorithm is 10% lower than ITA algorithm, 14.28% lower than SPUA algorithm, and 6.89% lower than MPSO algorithm.

In Fig. 8(a), as area expands, so do the number of UAVs. This is because the irregular distribution of users is more dispersed when the area increases and the number of users remains the same. Then, we need more UAVs to deploy. Fig. 8(b) shows that the total throughput of the terminal equipment gradually rises with the growth of the hotspot area length. This is because as the region grows, more UAVs are needed to assure user connection. Each UAV may serve fewer users while still maintaining the QoS of users, which improves the communication quality of users to some extent. Therefore, the throughput of the whole system is slowly increasing. Fig. 8(c) reveals there is no discernible trend in the total path loss of the user as the area length increases. This is because the expansion of the disaster area and the increase of the number of UAVs make the communication environment between users and UAVs has little change, so the change of hotspot area length have little impact on the total path loss. The reason for the above experimental results is that the ITA algorithm does not consider the number of UAVs. The SPUA algorithm When deploying UAVs is intended to reduce the number of UAVs, but the heuristic method used to solve it offers greater deployment benefits. The MPSO algorithm additionally considers the UAV's transmission power whiling getting a solution, which will result in performance loss. When the area is $1000 \times 1000 m^2$, the total path loss of the proposed algorithm is 11.32% lower than ITA algorithm, 15.86% lower than SPUA algorithm, and 5.04% lower than MPSO algorithm.

In Fig. 9(a), the number of deployed UAVs decreases as the available bandwidth of the UAV increases. This is because with the increase in the available bandwidth of UAVs, UAV coverage can serve more users under the premise of satisfying the QoS of users, thus reducing the number of UAVs. In Fig. 9(b), the total throughput of the terminal equipment increases when the UAV's bandwidth capacity expands. This is because that the number of users who can be served within each UAV's coverage increases as the UAV available bandwidth increases, and the amount of data the UAV send per unit time likewise rises. This factor is higher than the factor that the number of UAVs decreases, so the total system throughput will increase. According to Fig. 9(c), the total path loss of various UAV deployment strategies significantly lowers as available bandwidth rises. This is attributable to the fact that as UAV available bandwidth increases, the number of UAV

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

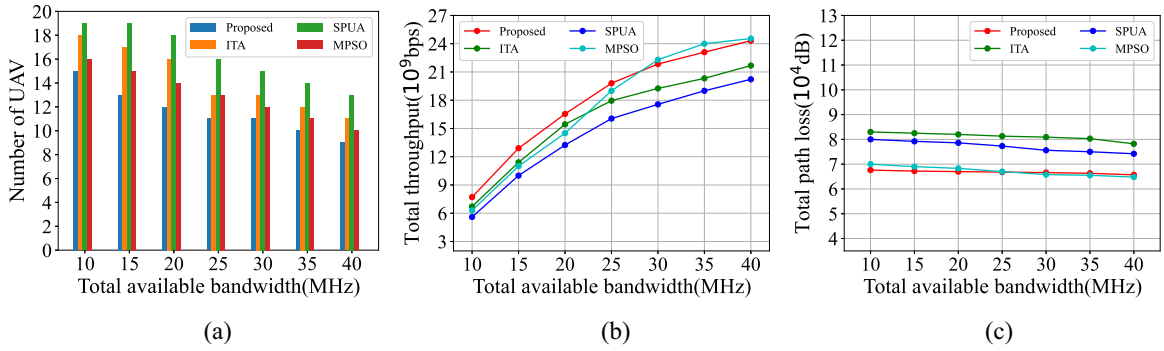


Fig. 9. Effect of available bandwidth of UAVs on experimental results. (a) The number of UAVs. (b) The total throughput. (c) The total path loss.

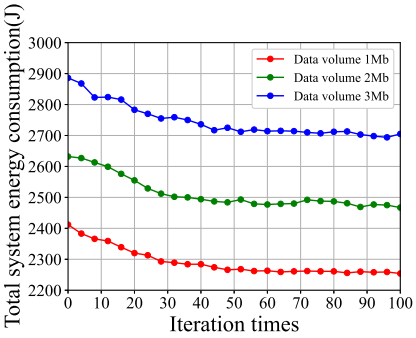


Fig. 10. Effect of iteration times on experimental results.

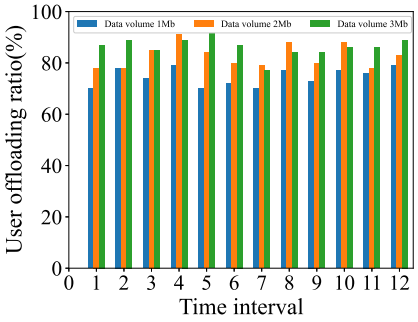


Fig. 11. User offloading ratio.

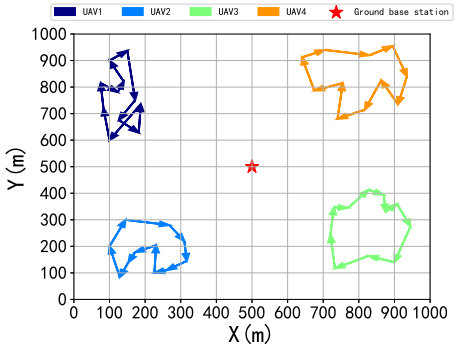


Fig. 12. UAV path planning.

deployments also increases, bringing more service users closer to the center of the UAV and reducing total path loss. When the available bandwidth of UAV is 25 MHz, the total path loss of the proposed algorithm is 18.55% lower than ITA algorithm, 15.15% lower than SPUA algorithm, and 2.61% lower than MPSO algorithm.

2) Experimental Verification of Computation Offloading Strategy: Fig. 10 shows the relationship between the total system energy consumption and the iteration times under different data volume tasks. From Fig. 10, the proposed algorithm in this paper tends to be stable after about 40 iterations, reaching convergence. After convergence, the amount of data is from 1 to 3 Mb, and the total energy consumption is reduced by 11.2%, 15.6% and 17.4% compared with the initial system energy consumption.

The user offloading ratio of the proposed computational offloading strategy algorithm at each time interval is shown in Fig. 11. This experiment sets up three groups of user offloading experiments with data volumes of 1 Mb, 2 Mb and 3 Mb. From Fig. 11, we find that the proportion of users who choose to offload is higher than 70%, and the larger the data volume, the more users choose to offload.

We set the experimental area as a square area of $1000 \times 1000m^2$, and the location of the GBS is (500, 500). Due to the user's mobility, the UAV needs to adjust its position according to the user's position and energy consumption requirements at each time slot. Therefore, Fig. 12 shows the trajectory of UAVs in the offloading period. First, the proposed UAV deployment method determines the number of UAVs deployed as well as the initial three-dimensional location. Because the beginning location and finishing position of the UAV are set to the same position in this paper, the trajectory of UAVs is end-to-end.

Fig. 13(a) shows that as the offloading task size increases, the system energy consumption for different offloading strategies increases. The proposed algorithm uses less energy than the TOA algorithm when the offload task size is less than 2.5 Mb, but more energy than the TOA algorithm when the task size exceeds 2.5 Mb. This is due to the poor convergence performance of the TOA algorithm during the entire iterative solution process. When the size of a task is 1.0 MB, the energy consumption of the proposed algorithm is 13.93% lower than TOA algorithm, 16.45% lower than MUT algorithm, and 18.28% lower than HJPQ algorithm. In Fig. 13(b), as the user offloading task size increases, so does the total system delay, and in terms of total system delay. This is because the proposed algorithm and the MUT algorithm simultaneously

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

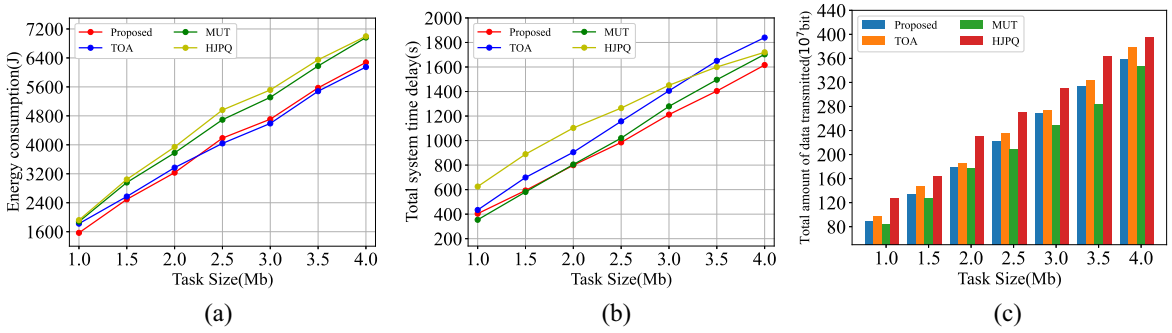


Fig. 13. Effect of offloading task size. (a) Total system energy consumption. (b)Total system delay. (c) Total amount of data transmitted.

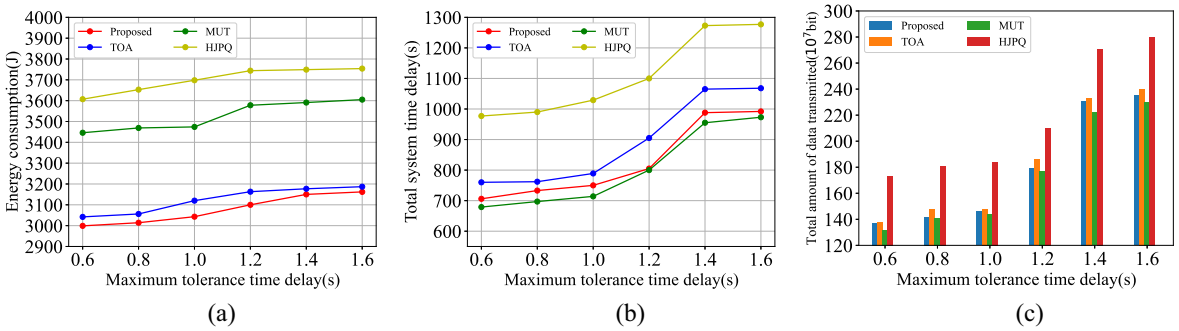


Fig. 14. Effect of user's maximum tolerant delay. (a) Total system energy consumption. (b) Total system delay. (c) Total amount of data transmitted.

optimize the deployment of UAVs, which is a significant determinant of the task offloading process's time consumption. Due to the proposed UAVs deployment method before, the total system delay is less than the MUT algorithm for a certain amount of data. when dealing with tasks involving large amounts of data, the MUT algorithm has a better transmission delay. In Fig. 13(c), the total transmission data volume of the proposed algorithm is barely distinguishable from the TOA algorithm and the MUT algorithm, but it is lower than the HJPQ algorithm. The primary cause is the HJPQ algorithm mainly considers the transmission rate when offloading and improves the user's QoS as much as possible. The MUT algorithm considers the total time of task completion.

Fig. 14(a) shows that when the user's maximum tolerated delay grows, the system energy consumption for different offloading strategies increases slightly before remaining flat. This is due to the fact that as the maximum tolerable delay rises, the user's task will progressively switch from being offloaded to the edge server to the local terminal. The proposed algorithm is not much different from the TOA algorithm in terms of energy consumption and outperforms the MUT algorithm and the HJPQ algorithm. The reason is the proposed algorithm and TOA algorithm are based on the minimum energy consumption as a goal. When the user's maximum tolerance delay is 1s, the energy consumption of the proposed algorithm is 2.47% lower than TOA algorithm, 12.41% lower than MUT algorithm, and 17.71% lower than HJPQ algorithm. In Fig. 14(b), the proposed algorithm is weaker than the MUT algorithm in terms of the total system delay, and outperforms the HJPQ and TOA algorithms. This is because the MUT algorithm aims to reduce the time required to complete an offloading operation. Additionally, the proposed algorithm outperforms the TOA algorithm in terms of UAV deployment. From Fig. 14(c), when the maximum tolerable delay of users increases, the total transmission data of systems with different offloading strategies remains stable before 1.2 s and increases after 1.2 s. This is because most tasks can be offloaded locally to meet their delay requirements as time goes by, and tasks will be offloaded from local computing to edge servers gradually. When the delay changes from 1.4 s to 1.6 s, it has little effect on the offloading decision of the user.

In Fig. 15(a), since the proposed algorithm takes energy consumption as the optimization goal, most tasks cannot guarantee the delay requirement when turning to local computing, so that the offloading decision will not change. Therefore, the proposed algorithm increases first and then tends to be gentle in energy consumption. The TOA algorithm and the proposed algorithm are aimed at minimizing energy consumption, so the two offloading strategies outperform the other two algorithms. When the computing power of UAV is 5 GHz, the energy consumption of the proposed algorithm is basically the same as that of TOA algorithm, 5.17% lower than MUT algorithm, and 10.53% lower than HJPQ algorithm. From Fig. 15(b), as the computing power of UAVs increases, the time required to complete the task decreases. And the MUT algorithm outperforms the other three algorithms in total system delay because it uses the minimum task completion time as its optimization goal. In addition, the proposed algorithm accurately deploys UAVs based on predicted user locations, which leads to a smaller transmission distance when the terminal equipment transmits data, reducing the time consumption in the data transmission stage. Therefore, compared with the TOA algorithm and the

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

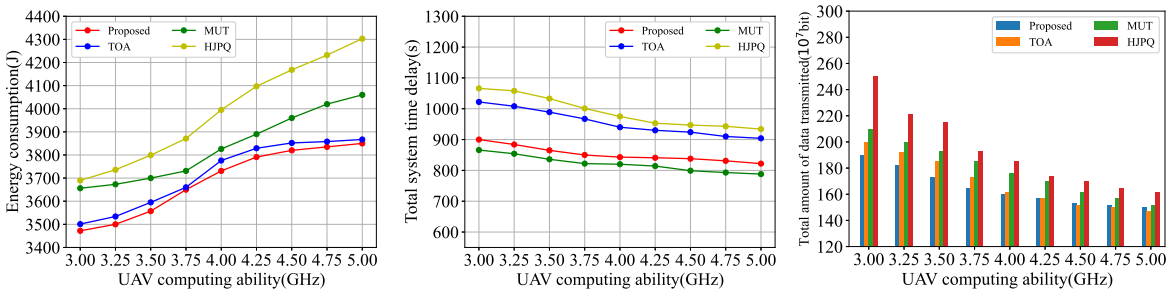


Fig. 15. Effect of computing ability of UAV. (a) Total system energy consumption. (b) Total system delay. (c) Total amount of data transmitted.

HJPQ algorithm, the proposed algorithm has a lower total system delay. In Fig. 15(c), the total amount of data transmitted by different algorithms in completing all tasks decreases with the increase of UAV computing power, and finally tends to be flat. This is because the UAV's computational energy consumption during the task offloading phase will rise as its computing ability grows, and different offloading algorithms have varying constraints on energy use. The decision to offload some tasks will move from offloading to edge servers to local computing, which minimizes the transmission of data in the system.

V. Conclusion and Future Work

In this paper, we design a UAV deployment scheme that can utilize users' QoS to determine the number and locations of UAVs. Then, a joint optimization problem comprising user association strategy, computing resource allocation strategy, and UAV trajectory is formulated to minimize the energy consumption for processing all terminal tasks. To solve this problem, the original problem is divided into three subproblems to be solved alternately by using the BCD method. The experiments show that the proposed strategy can reduce the path loss and the computation energy consumption.

In future work, we will minimize the number of required UAVs based on the given QoS of users. And we will consider the case of task partial offloading and use the deep reinforcement learning to solve the problem. Those work will be both technically challenging and extremely interesting.

Acknowledgment

The authors would like to thank the anonymous reviewers and editor for their valuable comments and suggestions.

References

[1] A. Musaddiq, R. Ali, S. W. Kim, and D.-S. Kim, "Learning-based resource management for low-power and lossy IoT networks," *IEEE Internet Things J.*, vol. 9, no. 17, pp. 16006–16016, Sep. 2022.

[2] L. Gu, M. Cui, L. Xu, and X. Xu, "Collaborative offloading method for digital twin empowered cloud edge computing on Internet of Vehicles," *Tsinghua Sci. Technol.*, vol. 28, no. 3, pp. 433–451, Jun. 2023.

[3] J. Cao, W. Feng, N. Ge, and J. Lu, "Delay characterization of mobile-edge computing for 6G time-sensitive services," *IEEE Internet Things J.* , vol. 8, no. 5, pp. 3758–3773, Mar. 2021.

[4] Z. Ding, D. Xu, R. Schober, and H. V. Poor, "Hybrid NOMA offloading in multi-user MEC networks," *IEEE Trans. Wireless Commun.*, vol. 21, no. 7, pp. 5377–5391, Jul. 2022.

[5] Y. C. Hu, M. Patel, D. Sabella, N. Sprecher, and V. Young, "Mobile edge computing—A key technology towards 5G," ETSI, Sophia Antipolis, France, White Paper, 2015.

[6] G. Qu, H. Wu, R. Li, and P. Jiao, "DMRO: A deep meta reinforcement learning-based task offloading framework for edge-cloud computing," *IEEE Trans. Netw. Service Manag.*, vol. 18, no. 3, pp. 3448–3459, Sep. 2021.

[7] A. M. Seid, G. O. Boateng, B. Mareri, G. Sun, and W. Jiang, "Multi-agent DRL for task offloading and resource allocation in multi-UAV enabled IoT edge network," *IEEE Trans. Netw. Service Manag.*, vol. 18, no. 4, pp. 4531–4547, Dec. 2021.

[8] Y. Gan and Y. He, "Trajectory optimization and computing offloading strategy in UAV-assisted MEC system," in *Proc. Comput. Commun. IoT Appl. (ComComAp)*, 2021, pp. 132–137.

[9] B. Liu, Y. Wan, F. Zhou, Q. Wu, and R. Q. Hu, "Resource allocation and trajectory design for MISO UAV-assisted MEC networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 5, pp. 4933–4948, May 2022.

[10] L. Zhang and N. Ansari, "On the number and 3-D Placement of in-band full-duplex enabled drone-mounted base-stations," *IEEE Wireless Commun. Lett.*, vol. 8, no. 1, pp. 221–224, Feb. 2019.

[11] Y. Wang, H. Wang, and X. Wei, "Energy-efficient UAV deployment and task scheduling in multi-UAV edge computing," in *Proc. Int. Conf. Wireless Commun. Signal Process. (WCSP)*, Nanjing, China, 2020, pp. 1147–1152.

[12] L. Wang, H. Zhang, S. Guo, and D. Yuan, "Deployment and association of multiple UAVs in UAV-assisted cellular networks with the knowledge of statistical user position," *IEEE Trans. Wireless Commun.*, vol. 21, no. 8, pp. 6553–6567, Aug. 2022.

[13] J. Luo, J. Song, F.-C. Zheng, L. Gao, and T. Wang, "User-centric UAV deployment and content placement in cache-enabled multi-UAV networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 5, pp. 5656–5660, May 2022.

[14] S. Zhang and N. Ansari, "3D drone base station placement and resource allocation with FSO-based backhaul in hotspots," *IEEE Trans. Veh. Technol.*, vol. 69, no. 3, pp. 3322–3329, Mar. 2020.

[15] M. Alzenad, A. El-Keyi, F. Lagum, and H. Yanikomeroglu, "3-D placement of an unmanned aerial vehicle base station (UAV-BS) for energy-efficient maximal coverage," *IEEE Wireless Commun. Lett.*, vol. 6, no. 4, pp. 434–437, Aug. 2017.

[16] I. Donevski, J. J. Nielsen, and P. Popovski, "Standalone deployment of a dynamic drone cell for wireless connectivity of two services," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Nanjing, China, 2021, pp. 1–7.

[17] T. Akram, M. Awais, R. Naqvi, A. Ahmed, and M. Naeem, "Multicriteria UAV base stations placement for disaster management," *IEEE Syst. J.*, vol. 14, no. 3, pp. 3475–3482, Sep. 2020.

[18] Y. Chen, H. Wang, F. Xiong, A. Li, Z. Qin, and J. Yuan, "Relay selection based on trajectory prediction for UAV networks," in *Proc. Int. Conf. Wireless Commun. Signal Process. (WCSP)*, Nanjing, China, 2020, pp. 1212–1217.

[19] L. Lyu, F. Zeng, Z. Xiao, C. Zhang, H. Jiang, and V. Havyarimana, "Computation bits maximization in UAV-enabled mobile-edge computing system," *IEEE Internet Things J.*, vol. 9, no. 13, pp. 10640–10651, Jul. 2022.

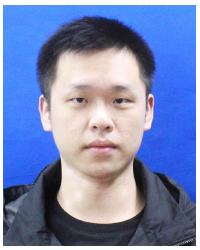
[20] N. Zhao, Z. Ye, Y. Pei, Y.-C. Liang, and D. Niyato, "Multi-agent deep reinforcement learning for task offloading in UAV-assisted mobile edge computing," *IEEE Trans. Wireless Commun.*, vol. 21, no. 9, pp. 6949–6960, Sep. 2022.

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.

[21] S. Zhu, L. Gui, J. Chen, Q. Zhang, and N. Zhang, "Cooperative computation offloading for UAVs: A joint radio and computing resource allocation approach," in *Proc. IEEE Int. Conf. Edge Comput. (EDGE)*, San Francisco, CA, USA, 2018, pp. 74–79.
[22] M. Dai, Z. Su, J. Li, and J. Zhou, "An energy-efficient edge offloading scheme for UAV-assisted Internet of Things," in *Proc. IEEE 40th Int. Conf. Distrib. Comput. Syst. (ICDCS)*, Singapore, 2020, pp. 1293–1297.
[23] J. Xiong, H. Guo, and J. Liu, "Task offloading in UAV-aided edge computing: Bit allocation and trajectory optimization," *IEEE Commun. Lett.*, vol. 23, no. 3, pp. 538–541, Mar. 2019.
[24] A. Gao, Q. Wang, Y. Hu, and W. Duan, "An offloading optimization scheme for multi-UAV aided network in mobile computing," in *Proc. Int. Wireless Commun. Mobile Comput. (IWCMC)*, Limassol, Cyprus, 2020, pp. 1468–1473.
[25] S. Sun, G. Zhang, H. Mei, K. Wang, and K. Yang, "Optimizing Multi-UAV deployment in 3-D space to minimize task completion time in UAV-enabled mobile edge computing systems," *IEEE Commun. Lett.*, vol. 25, no. 2, pp. 579–583, Feb. 2021.
[26] A. Al-Hourani, S. Kandeepan, and S. Lardner, "Optimal LAP altitude for maximum coverage," *IEEE Wireless Commun. Lett.*, vol. 3, no. 6, pp. 569–572, Dec. 2014.
[27] L. Zhang and N. Ansari, "Latency-aware IoT service provisioning in UAV-aided mobile-edge computing networks," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 10573–10580, Oct. 2020.
[28] L. R. Rabiner, "A tutorial on hidden Markov models and selected applications in speech recognition," *Proc. IEEE*, vol. 77, no. 2, pp. 257–286, Feb. 1989.
[29] S. Wan, J. Lu, P. Fan, and K. B. Letaief, "Toward big data processing in IoT: Path planning and resource management of UAV base stations in mobile-edge computing system," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 5995–6009, Jul. 2020.
[30] N. N. Ei, S. W. Kang, M. Alsenwi, Y. K. Tun, and C. S. Hong, "Multi-UAV-assisted MEC system: Joint association and resource management framework," in *Proc. Int. Conf. Inf. Netw. (ICOIN)*, 2021, pp. 213–218.
[31] J. Wang, C. Jin, Q. Tang, N. N. Xiong, and G. Srivastava, "Intelligent ubiquitous network accessibility for wireless-powered MEC in UAV-assisted B5G," *IEEE Trans. Netw. Sci. Eng.*, vol. 8, no. 4, pp. 2801–2813, Oct.-Dec. 2021.
[32] M. D. Nguyen, L. B. Le, and A. Girard, "Integrated UAV trajectory control and resource allocation for UAV-based wireless networks with co-channel interference management," *IEEE Internet Things J.*, vol. 9, no. 14, pp. 12754–12769, Jul. 2022.
[33] Y. Zheng, X. Xie, and W.-Y. Ma, "GeoLife: A collaborative social networking service among user, location and trajectory," *IEEE Data Eng. Bull.*, vol. 33, no. 2, pp. 32–39, Jun. 2010.
[34] G. B. Huang, M. Ramesh, T. Berg, and E. Learned-Miller "Labeled faces in the wild: A database forstudying face recognition in unconstrained environments," in *Proc. Workshop Faces Real-Life Images Detect. Alignment Recognit.*, 2008, pp. 1–11.



Chunlin Li received the B.S. degree and M.Sc. degree in computer science from the Wuhan University of Technology (WUT), China, in 1996 and 2000, respectively, and the Ph.D. degree in computer software and theory from the Huazhong University of Science and Technology, Wuhan, China, in 2003. She is currently a Professor and a Ph.D. Tutor with the School of Computer and Artificial Intelligence, WUT. She has published over 60 technical papers and obtained over 20 authorized invention patents. Her research interests include high performance computing, cloud/edge computing, distributed optimization, UAV communication, and Internet of Things. She has received multiple scientific awards at or above the provincial and ministerial levels. She won the Microsoft Fellowship Award and the IBM Global Ph.D. Elite Program Scholarship Award in 2003. She was selected for the New Century Excellent Talent Program of Ministry of Education (MOE), and the High End Talent Leading Program of Hubei Province, in 2008 and 2012, respectively. She was selected on the World Top 2% Scientists List in 2021 and 2022. She also serves as an Expert for reviewing Key Projects of the National Natural Science Foundation, National Outstanding Youth Fund, the National Key Research and Development Program, the Talent Program of the Ministry of Science and Technology, the Yangtze River Scholars (MOE), and the National Science and Technology Awards, and a reviewer of multiple IEEE/ACM Transaction/Journal papers.



Yongzheng Gan received the B.S. degree in information and computing science from the Wuhan University of Science and Technology in 2017. He is currently pursuing the M.S. degree with the School of Computer and Artificial Intelligence, Wuhan University of Technology. His research interests include cloud computing and artificial intelligence.



Yong Zhang received the M.S. degree from Tiangong University in 2020. He is currently pursuing the Ph.D. degree with the School of Computer and Artificial Intelligence, Wuhan University of Technology. His research interests include cloud computing and artificial intelligence.



Youlong Luo received the M.S. degree in telecommunication and system and the Ph.D. degree in finance from the Wuhan University of Technology in 2003 and 2012, respectively, where he is a Vice-Professor of Management. His research interests include cloud computing and electronic commerce.

Authorized licensed use limited to: Indian Institute of Petroleum and Energy. Downloaded on October 26,2026 at 11:55:51 UTC from IEEE Xplore. Restrictions apply.